

Assignment 3 - Practice Quiz A: explanations

1. Answer: 7

Only three unions join distinct components; the repeated union and self-union do not change the count.

2. Answer: 8

The new size is the sum of the two distinct component sizes.

3. Answer: $O(n)$ for that single find

Distinguish initialization, amortized sequence cost, memory, and a single operation in a stated parent structure.

4. Answer: 13

The two lightest edges form a spanning tree; a third edge creates a cycle.

5. Answer: 24

All three edges are bridges in their components; isolated vertices need no edges.

6. Answer: 8

A component with x vertices contributes $x-1$ edges; sum over all components.

7. Answer: 7

Prim chooses a minimum crossing edge; an internal edge cannot add a vertex.

8. Answer: 11

It chooses the first two edges. The optimum chooses the first and third edges, so the excess is the second weight minus the third.

9. Answer: 7

$\{0,1\}$ and $\{2,3\}$ are two SCCs; each other vertex is a singleton.

10. Answer: 3

Remove internal edges and deduplicate the component pairs $A \rightarrow B$, $B \rightarrow C$, $A \rightarrow C$.

11. Answer: The component graph is acyclic, but the original internal dependency cycle still needs resolution.

Condensation describes cycles; it does not make the original cyclic prerequisites executable.

12. Answer: 5

The second pass considers vertices in reverse finish-list order.

13. Answer: 4

Enumerate the orders respecting every edge. Unrelated vertices may appear in either order; count each full permutation once.

14. Answer: Report that no topological order exists.

An edge to a grey vertex closes a directed cycle, contradicting a topological ordering.

15. Answer: 16

Reverse postorder reverses the finish sequence.

16. Answer: $O(n)$

Count vertices and inspected edges or matrix cells; distinguish the input representation and auxiliary memory.

17. Answer: 2

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ costs $6 - (8) + 4 = 2$, less than the direct route through 2.

18. Answer: -1

The new route costs $7 - (8) = -1$. This is a reachable negative distance, not an unreachable marker.

19. Answer: 6

Follow the predecessor chain to the source, then reverse it. The number of edges is one less than the number of vertices.

20. Answer: 6000000000

Accumulate in a 64-bit integer; a 32-bit signed integer can overflow.